

Making AI-generated mathematics auditable with OPDP

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722

manuscripts assessed

372

result families

102,819

archived problem records

254

pages of analysis

THE AUTHOR'S CONTRIBUTION: OPDP

Alejandro Zarzuelo Urdiales created and authored the **Open Problem Difficulty Profile (OPDP)**, with **ChatGPT 5.6 Sol assistance**. The framework was further refined with newer and diverse models to capture the multidimensional complexity of mathematics. OPDP replaces opaque labels with explicit dimensions, public rationales, source identifiers and versioned assessments. Applied here to OpenAI's mathematics release, it gives every manuscript a description, a precise scoring target, an archive relation, a justified difficulty assessment and a conditional OpenMath judging calculation. It connects a large catalogue of claims to inspectable research judgments.

WHY THIS ANALYSIS MATTERS

The unit of mathematical progress is a scoped result, not a paper count. General theorems, conditional results, finite constructions, special cases and alternative proofs cannot be compared by volume alone. Hypotheses, quantifiers, conclusions and computational models determine whether two claims address the same target. OPDP makes these distinctions explicit before assigning credit. This supports expert-review prioritization, transparent competition judging and an auditable account of AI-assisted mathematical research for technical readers and scientific publishers.

A MULTIDIMENSIONAL MECHANISM

Nine normalized burdens feed the 0-1000 full-solution difficulty **D**: AI-relative reasoning (weight 400), intrinsic mathematical difficulty (170), human resistance (90), inverse tractability (150), verification (45), formalization (35), prerequisites (40), context/ambiguity (40) and tool constraints (30). Discovery, checking and formal encoding are therefore separate obstacles. For weighted burden **R**, the published transformation is $D = 1000[1 - (1 - R/1000)^{1.4}]$; new estimates are rounded to 25. Scores assume access to public literature and computational/proof tools, with a nominal 100-agent-hour ceiling. Unknown human attention is neutral (0.5), not invented historical effort.

FROM PROFILE TO JUDGING

Under the OpenMath non-focus scenario, $F = m p D^2/1000$: **m** distinguishes an original target from an admitted variation; **p** describes progress toward the explicitly named target. Completing a narrow result is not automatically completing its broader parent. This is a judging policy, not a measured law of scientific value. Manuscript-level scenarios must not be added as independent awards: companions, consequences and alternative proofs require family-level accounting and novelty review.

WHAT THE ANALYSIS ESTABLISHES

All 722 catalogue manuscripts were acquired and individually documented in a **254-page** report, across 372 upstream result families and 17 disciplines. Comparison with the frozen **102,819-record** OPDP atlas yields **59 same-target scope matches (8.2%)**, **91 partial or special-case relations (12.6%)**, **287 related targets (39.8%)**, **147 lexical candidates (20.4%)** and **138 unresolved matches (19.1%)**. These are manuscript-level scope assessments, not verified resolutions or mutually distinct mathematical accomplishments.

CONCRETE SAFEGUARDS AGAINST OVERCLAIMING

The reported asymptotic matrix-multiplication bound **omega <= 9/4** is a different target from a fixed 3 x 3 scheme using 23 products; the latter's registered difficulty is not transferred. Likewise, constant-error synthesis with a target-dependent Boolean oracle is not equivalent to exact gate synthesis in a different computational model. The report preserves **26** applicable original scores, supplies **692** new lowest-confidence (C0) estimates, and assigns no new-resolution credit to **four** acknowledged reconstructions of known main results. Alternative proofs may still have scientific value; they are not new resolutions of the already-settled target.

VALUE FOR AI RESEARCH AND SELF-IMPROVEMENT

A model may search effectively yet fail at long dependencies, literature integration, independent verification or formalization. Separating these barriers can inform problem selection, tool development and allocation of expert review. For recursive self-improvement (RSI), OPDP offers a way to specify which bottleneck an intervention is intended to improve; this analysis does **not** demonstrate RSI. A prospective test would freeze targets and budgets, rate difficulty before seeing outcomes, hold out source families, independently check results and compare success/cost by dimension. Source-stratified evaluation is essential to avoid confusing a change in problem mix with a change in capability.

EVIDENCE BOUNDARY AND REUSE

These are retrospective, source-based editorial judgments, not a calibrated AI success probability or proof certification. No solve campaign or Lean kernel replay was performed. The atlas includes excerpt-derived C0 records, not 102,819 independently verified distinct open conjectures. Unresolved matching proves neither novelty nor absence. Official new-target admission requires three independent difficulty assessments plus scope, novelty and proof review. The public JSON retains per-target rationale, scope, source hashes, component inputs and calculation traces. The analysis is published separately and does not silently modify canonical atlas records.

Read the complete evidence: [Full analysis \(254-page PDF\)](#) | [Scoring JSON](#) | [OPDP repository](#)

<https://github.com/alejandrozu/ulam-opdp-difficulty-atlas>

Source: [OpenAI release, 6 October 2026](#); openai/math snapshot adc7f1241b42; OPDP v1.8 snapshot e0e18186a142. Full hashes and individual justifications are in the linked report/JSON.